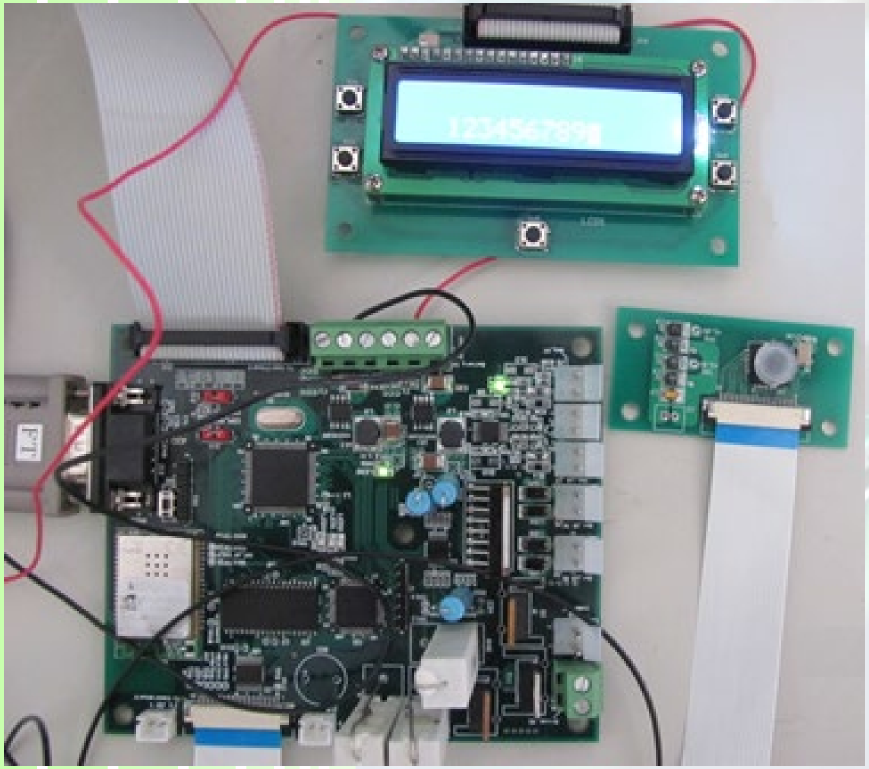
Embedded Engineer, Firmware Developer, PCB designer

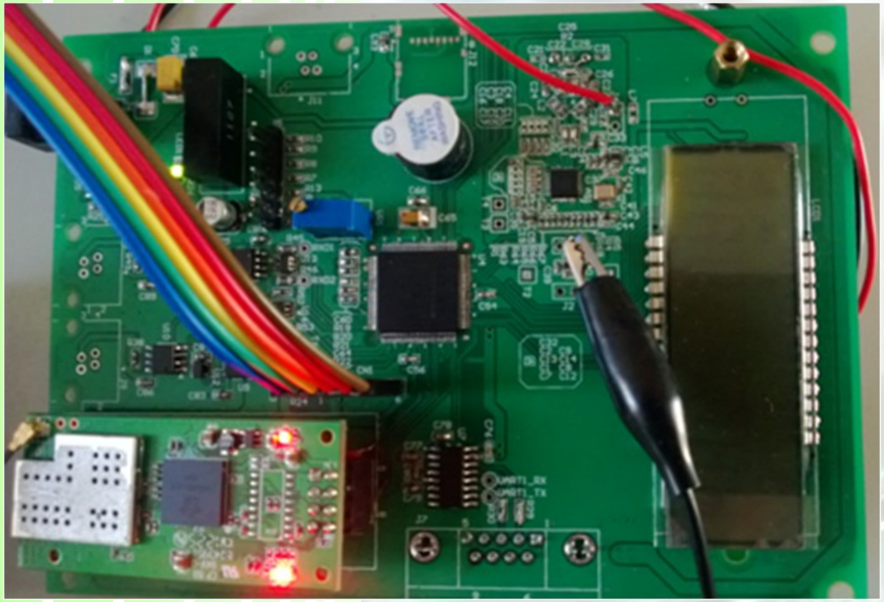
Ball Shoot Game Device



- Two full bridge motor control
- Two PWM motor control
- Camera and display control
- Image processing technique
- Communication with a PC or iPhone by Wi-Fi

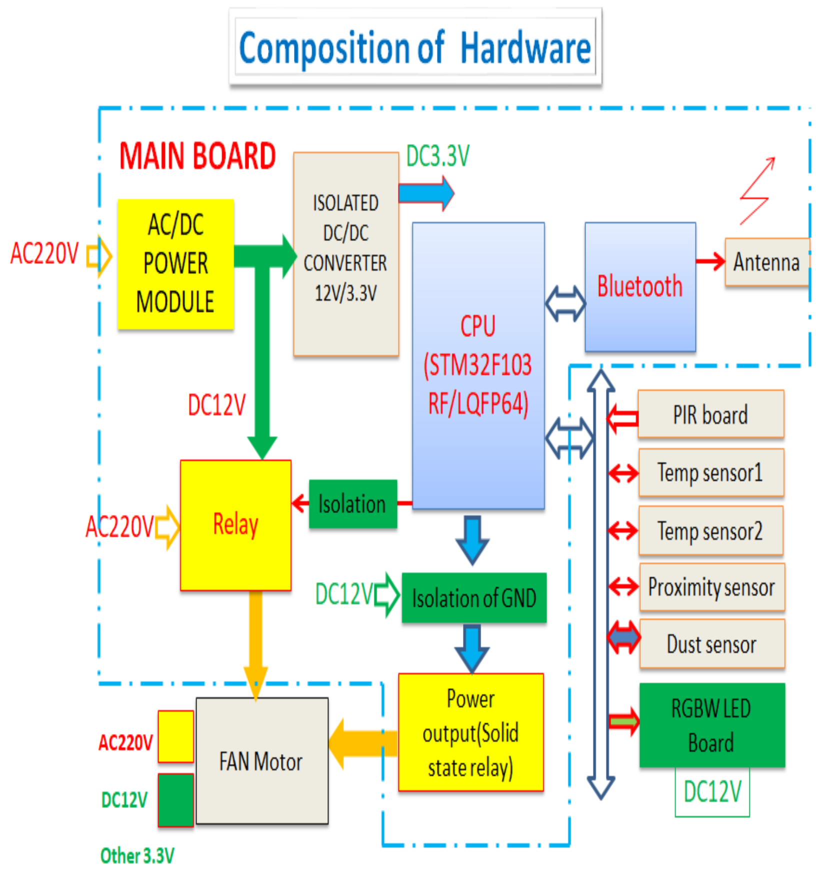
- dsPIC33EP256MU810, MRF24WB0MA/B, TCM8240MD camera chip

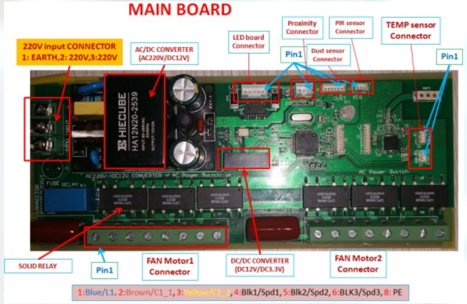
Power Payment System Controller

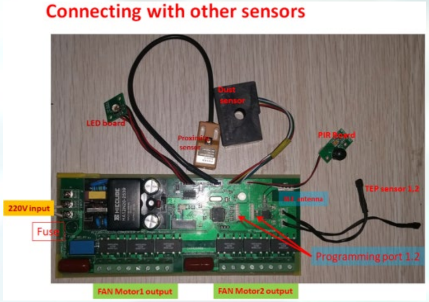


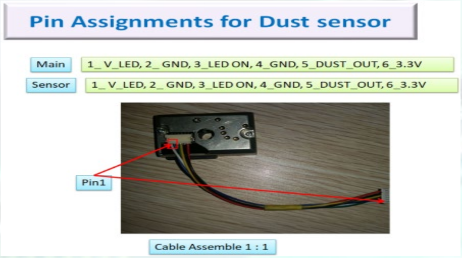
- PIC24FJ128GA310, Wi-Fi M03 module and AS3911 RFID Chip
- Reading data stored on RFID cards
- Data include power, voltage, current, cumulative energy consumption and remaining energy (balance)
- Customers can key-in as many tokens as they need through the Matrix Key-Pad
- The controller sends and receives data to and from the meter via RS485 communication
- The controller sends electrical parameters to the server via Wi-Fi communication

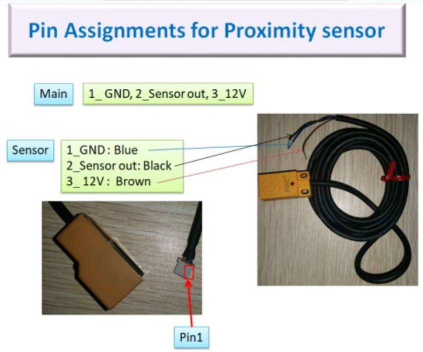
BigFAN-Motor driver

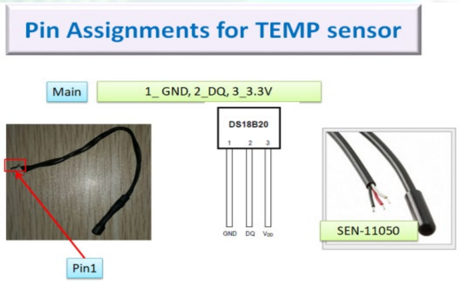


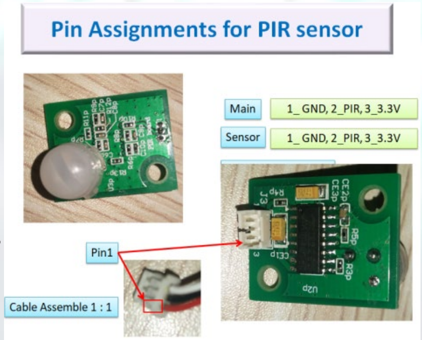






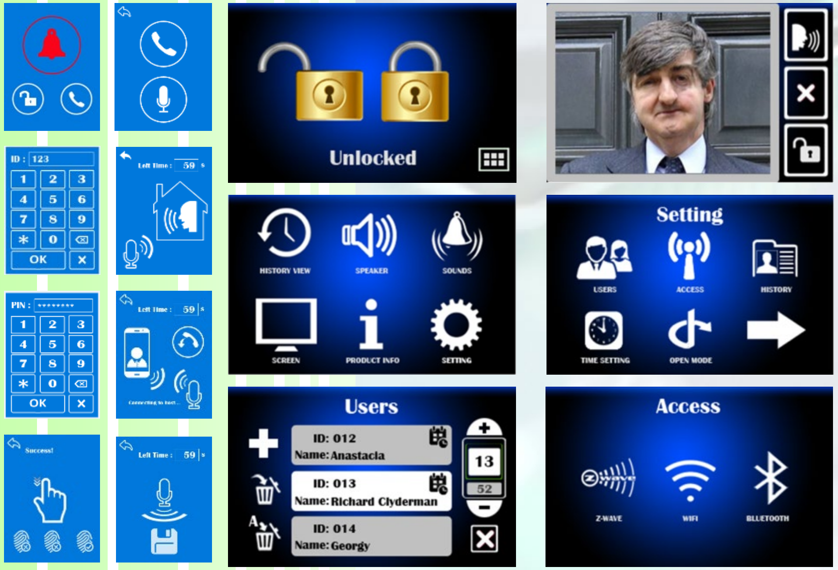






STM32F103RF/LQFP64, CC2540F256RHAT, S9009-B, LZ4-00MA00, GP2Y1010AUOF, DS18B20

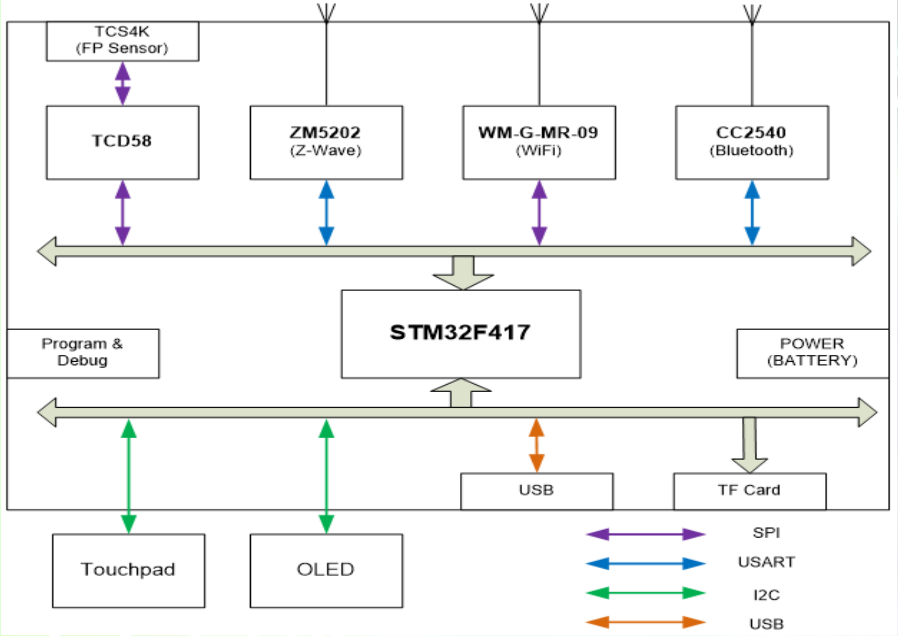
Smart door lock

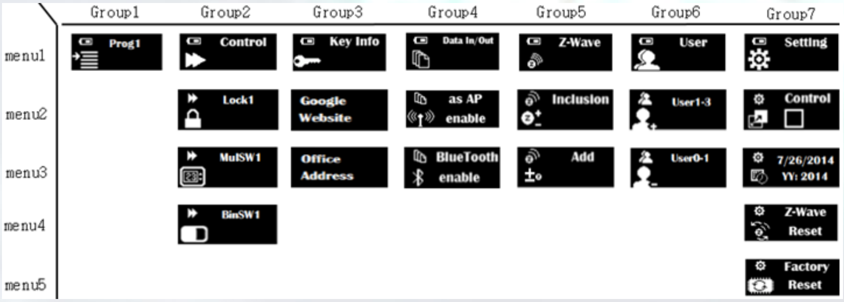


- PIR Sensor, Camera
- Pin Code, Fingerprint Scanner
- Q-Touch, Touch LCD
- MIC & Speaker
- 6-Axis Door Motion
- Battery System
- Wi-Fi, Z-wave, Bluetooth
- Voice Chatting
- Wireless conversation with a hand-phone
- STM32F417IGH6, ZM5202, WM-G-MR-09_REF2, CC2540F256RHAR, MC34018DW, VS1063a, TM2291, TCS4K / TCD58 chipset

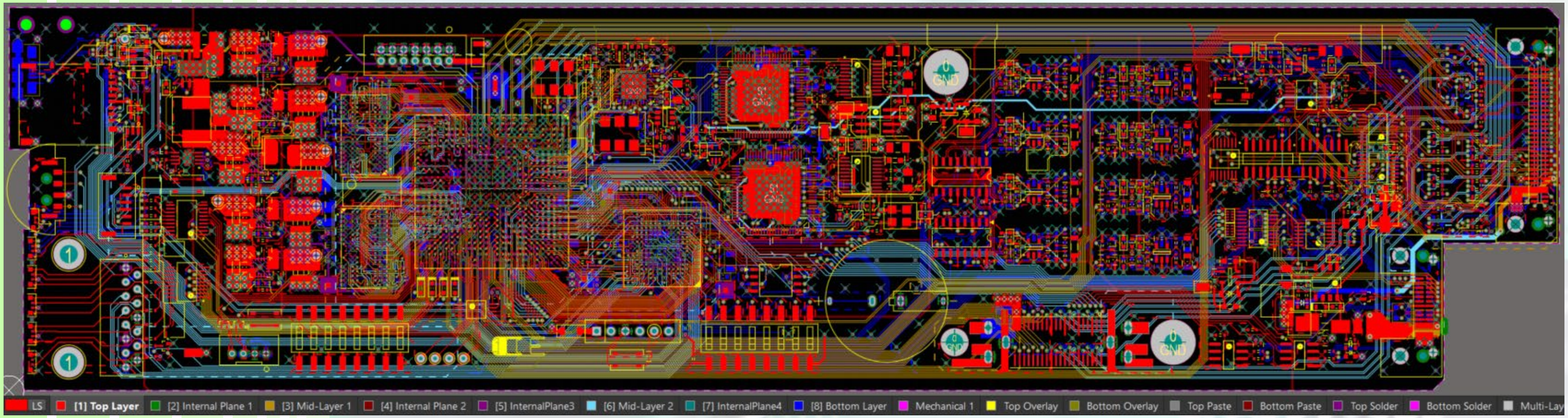
Smart Fob remote controller

- Pin Code, Fingerprint Scanner
- O-LED
- Touchpad
- Battery System
- Wi-Fi, Z-wave, Bluetooth
- Wireless conversation with a hand-phone
- STM32F417IGH6, TCS4K / TCD58 chipset, ZM5202, WM-G-MR-09_REF2, CC2540F256RHAR



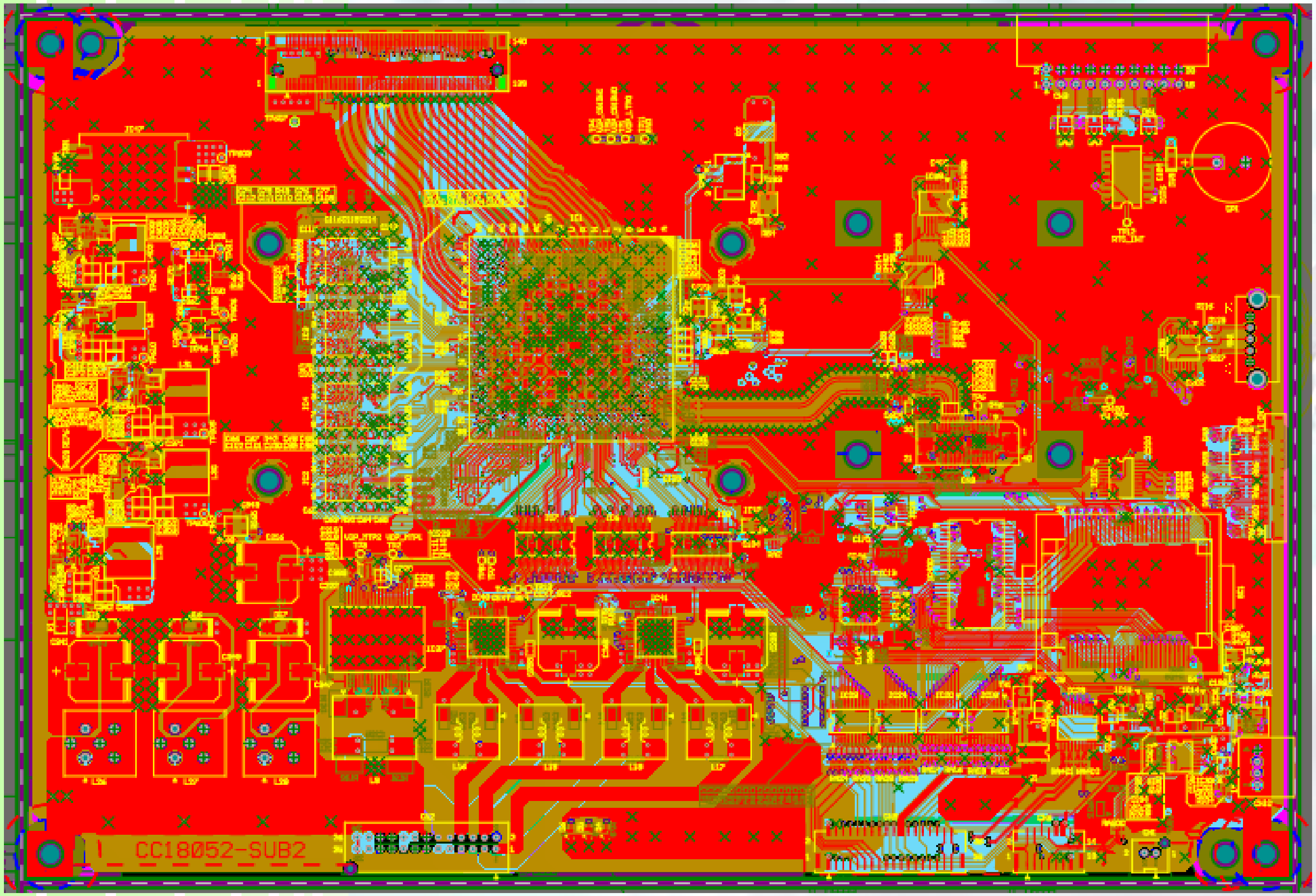


The PCB design of the Mainboard for the Audio and Video Signal Generator



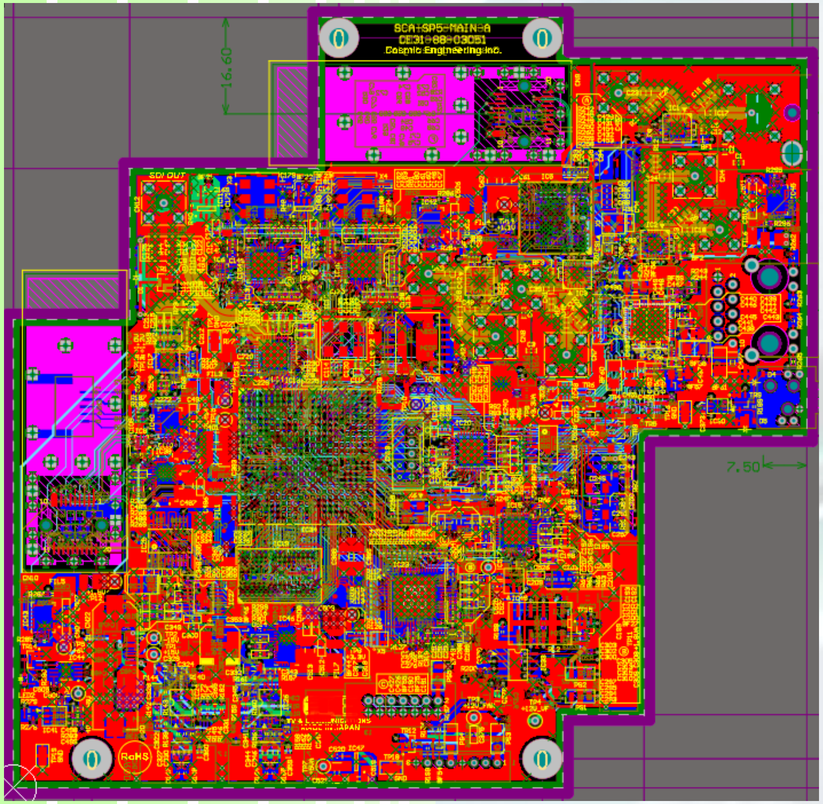
- 8Layers design
- Video Timing & status, PLL for GenCLK
- NTSC Decoder
- RS-485, RS-422, SDI, SPI, I2C, DDR3
- USB-FS OTG, microSD
- Analog and digital mixed
- STM32F746NGH6, XC7K160T-1FBG676C, MT41J128M16

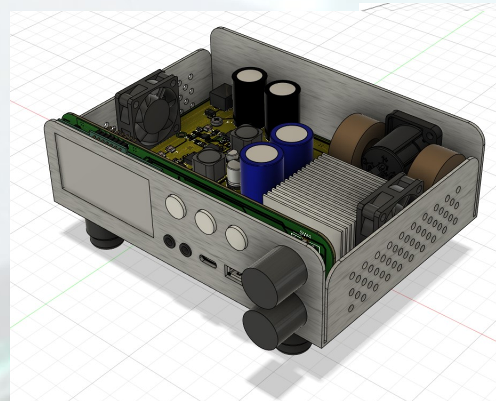
The PCB design of the Gen 6464-12G High-Speed Signal Generator Mainboard

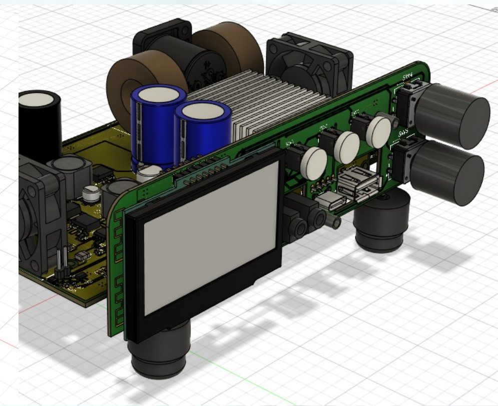


- 12Gbps high speed signal processing technique
- DDR4 interface and technique
- Analog and digital mixed
- M21080G-12, XCKU035-1FBVA676C, MT40A256M16GE

The PCB Design of Video & Audio Broadcasting System Mainboard



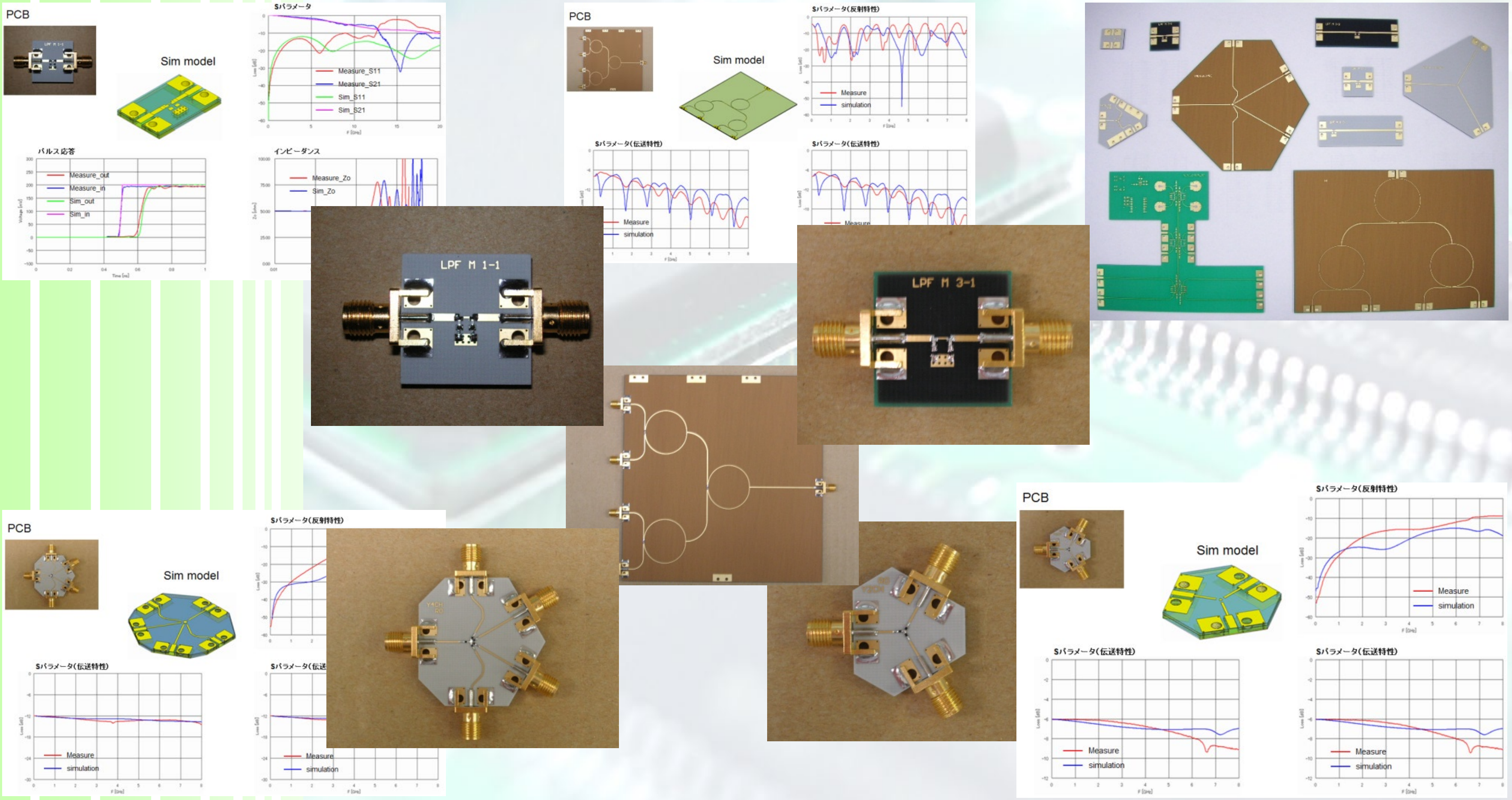




- Modulator and Demodulator technique, DDR4 interface
- Analog and digital mixed
- Control system using 7 FPGAs

- XCKU115-FLVA1517, XC6SLX45-3FGG676, MT40A256M16LY, GS1662-IBE3, GS1661A-IBE3, RTL8211E-VL

Simulation Application of DIB Design Model Parameter



Thank you very much for your attetion!